

Rescattering effects for a three-body final state

MIKHAIL MIKHASENKO

Thanks to:

Prof. Dr. BERNHARD KETZER, HISKP, Uni Bonn

Prof. Dr. ADAM SZCZEPANIAK, Uni Indiana.

HISKP, Uni Bonn
JPAC, Indiana-JLab

February 15th, 2016

Contents

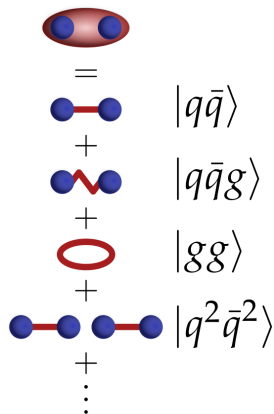
- 1 Motivation
 - Introduction
- 2 Importance of rescattering
 - Final state interaction
- 3 Rescattering equations
 - Unitarity
 - Khuri-Threiman formalism
 - Example
- 4 Conclusion

Interpretation of resonance

- Beyond bound quark-anti-quark states, other exotic states of QCD are possible
- Possible exotic states are
 - hybrids: $|q\bar{q}g\rangle$,
 - glueballs: $|gg\dots\rangle$,
 - multi-quark states:
 - tetraquarks $|q\bar{q}q\bar{q}\rangle$,
 - hadro-molecules $|(q\bar{q})(q\bar{q})\rangle$.
 - ...

Several candidates to be exotic states have been observed

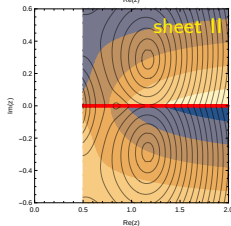
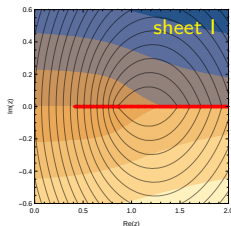
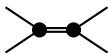
XYZ-states, $a_1(1420)$, $P_c(4450)$.



Complex structure

- Amplitude $t(s)$ of scattering process is complex function of s .
- Every resonance is a pole in the amplitude
- Every one channel for decay produces cut along the real axis

Two-body resonance



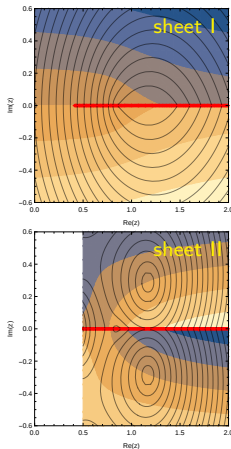
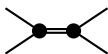
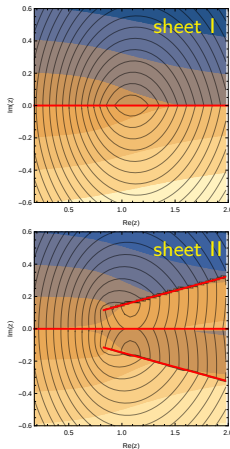
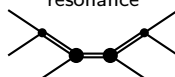
$t^{-1}(s)$ is shown in s -plane,
color code is $\text{Im}[t^{-1}(s)]$,
equipotential lines are
 $\text{ABS}[t^{-1}(s)]$.

Complex structure

- Amplitude $t(s)$ of scattering process is complex function of s .
- Every resonance is a pole in the amplitude
- Every one channel for decay produces cut along the real axis

$t^{-1}(s)$ is shown in s -plane,
color code is $\text{Im}[t^{-1}(s)]$,
equipotential lines are
 $\text{ABS}[t^{-1}(s)]$.

Two-body resonance

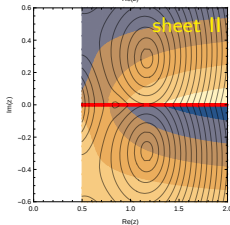
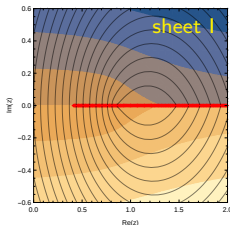
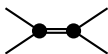
Quasi-two-body
resonance

Complex structure

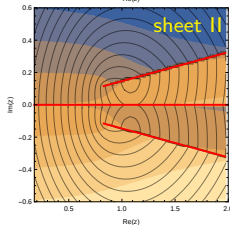
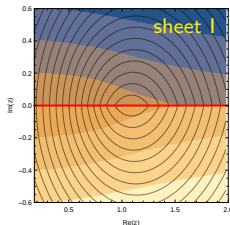
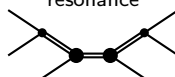
- Amplitude $t(s)$ of scattering process is complex function of s .
- Every resonance is a pole in the amplitude
- Every one channel for decay produces cut along the real axis

$t^{-1}(s)$ is shown in s -plane,
color code is $\text{Im}[t^{-1}(s)]$,
equipotential lines are
 $\text{Abs}[t^{-1}(s)]$.

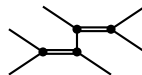
Two-body resonance



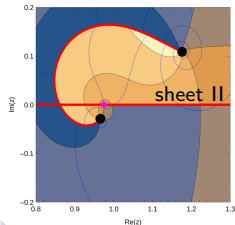
Quasi-two-body resonance



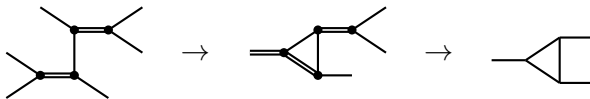
Cross-channel exchange



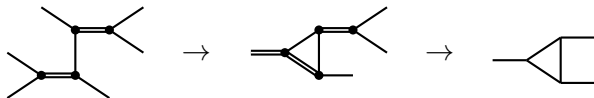
- Additional structure appears on the second sheet.
- log branching point can be close to the physical region.



Triangle diagram

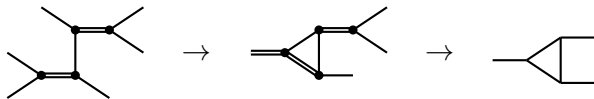


Triangle diagram



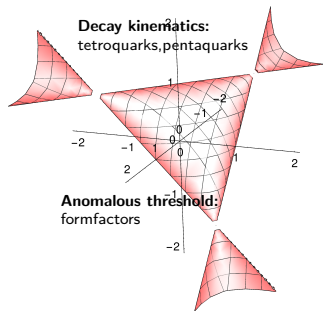
$$\begin{aligned}
& \text{Diagram: } p_0, s_0 \text{ splits into } k_1, m_1^2 \text{ and } k_2, m_2^2, \text{ which then recombine into } p_1, s_1 \text{ and } p_2, s_2. \\
& = g^3 \int \frac{d^4 k_1}{(2\pi)^4 i} \frac{1}{\Delta_1 \Delta_2 \Delta_3} = \frac{g^3}{16\pi^2} \int_0^1 \frac{dx dy dz}{D} \delta(1-x-y-z), \\
& D = x m_1^2 + y m_2^2 + z m_3^2 - x y s_0 - z x s_1 - y z s_2.
\end{aligned}$$

Triangle diagram



$$\begin{aligned}
 & \text{Triangle diagram with external momenta } p_0, s_0 \text{ and } p_1, s_1, p_2, s_2, \text{ and internal momenta } k_1, m_1^2, k_2, m_2^2, k_3, m_3^2 \\
 &= g^3 \int \frac{d^4 k_1}{(2\pi)^4 i} \frac{1}{\Delta_1 \Delta_2 \Delta_3} = \frac{g^3}{16\pi^2} \int_0^1 \frac{dx dy dz}{D} \delta(1 - x - y - z), \\
 & D = x m_1^2 + y m_2^2 + z m_3^2 - x y s_0 - z x s_1 - y z s_2.
 \end{aligned}$$

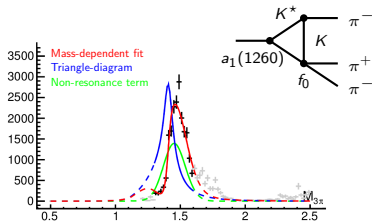
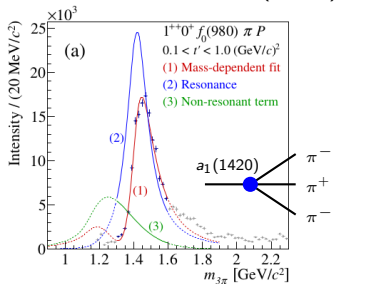
- $\Delta_i = m_i^2 - k_i^2$ is propagators of the particles in the loop,
- Positions of singularities are given by Landau equations. [Landau, Nucl. Phys. **13**, 181 (1959)]
- Landau surface is represented in normalized invariants (y_0, y_1, y_2) , $y_i = \frac{s_i - m_{i1}^2 - m_{i2}^2}{2m_{i1}m_{i2}}$



Examples

hypotheses for an explanation of exotics

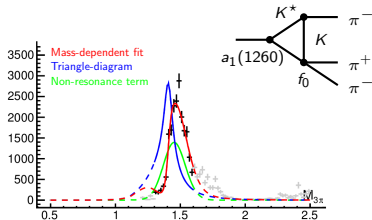
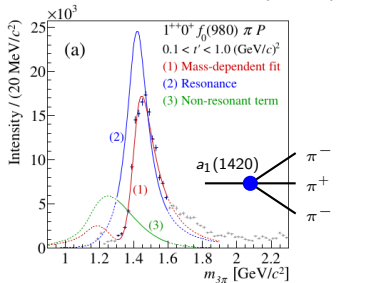
COMPASS: exotic $a_1(1420)$



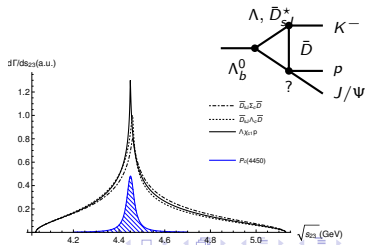
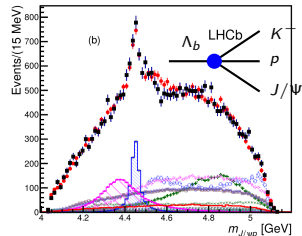
Examples

hypotheses for an explanation of exotics

COMPASS: exotic $a_1(1420)$



LHCb: pentaquark $P_c(4450)$

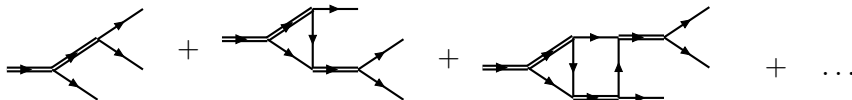


Rescattering series

How a further rescattering changes an amplitude?

- Isobar model is common tool of the data analysis (PWA at VES, COMPASS, BES, CLEO, LHCb, ...).
- 1th order rescattering is triangle diagram. One loop, four integrals can be reduced to one.
- 2th is two-loop diagram. Very complicated to evaluate!

Isobar model + 1th-rescattering + 2th-rescattering + etc ...



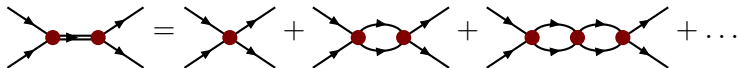
There is alternative method.

Two body unitarity

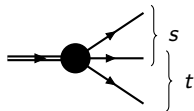
Unitarity = propability conservation. $\hat{S} \cdot \hat{S}^\dagger = \mathbb{I}$.

$$\hat{S} = \hat{\mathbb{I}} + i\hat{T} \Rightarrow \hat{T} - \hat{T}^\dagger = i\hat{T}\hat{T}^\dagger, \Rightarrow \Delta T = iT_\rho T^*.$$

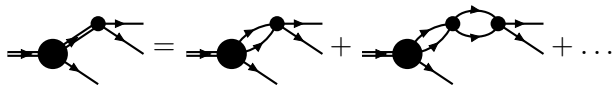
Two-body unitarity and resonance



$$T = \frac{g^2}{m^2 - s - im\Gamma} = \frac{1}{K^{-1} - i\rho} = K + K i\rho K + K i\rho K i\rho K + \dots, \quad K = \frac{g^2}{m^2 - s}.$$



Subchannel resonances



Complete only if interaction with particle 3 is negligible

Formalism

Khuri-Treiman equation

$$A = A^{(s)} + A^{(t)} + A^{(u)}, \quad A^{(j)} = \sum_l (2l+1) a_l^{(j)}(s_j) P_l(\cos \theta^{(j)})$$

Due to unitarity: $a_l^{(s)} = \underbrace{t_l^{(s)} c_l^{(s)}}_{\text{Isobar model}} + \underbrace{\frac{t_l^{(s)}}{2\pi} \int \frac{b_l^{(s)} \rho}{s' - s} ds'}_{\text{rescattering corrections}}.$

- $a_l^{(s)}$ is a corrected two-body amplitude,
- $b_l^{(s)}$ is a projection of cross channel waves.

$$b_l^{(i)} = \hat{M}_{ij} b_{l'}^{(j)} + c H_l^{(i)}$$

Formalism

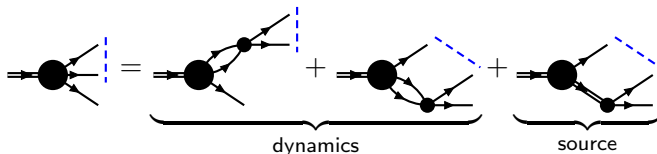
Khuri-Treiman equation

$$A = A^{(s)} + A^{(t)} + A^{(u)}, \quad A^{(j)} = \sum_l (2l+1) a_l^{(j)}(s_j) P_l(\cos \theta^{(j)})$$

Due to unitarity: $a_l^{(s)} = \underbrace{t_l^{(s)} c_l^{(s)}}_{\text{Isobar model}} + \underbrace{\frac{t_l^{(s)}}{2\pi} \int \frac{b_l^{(s)} \rho}{s' - s} ds'}_{\text{rescattering corrections}}.$

- $a_l^{(s)}$ is a corrected two-body amplitude,
- $b_l^{(s)}$ is a projection of cross channel waves.

$$b_l^{(i)} = \hat{M}_{ij} b_{l'}^{(j)} + c H_l^{(i)}$$



Input/Output

Model is given by

- Set of waves up to L_j for every channel s, t, u .

Example $\pi^- \pi^+ \pi^-$: s,t-channel isobars are $\sigma, f_0(L=0), \rho(L=1)$

- Parameterizations of elastic amplitudes $t_l^{(j)}$

Example $\pi^- \pi^+ \pi^-$: $(\pi\pi)_S$ -wave, $(\pi\pi)_P$ -wave from K-function/phase shift

Input/Output

Model is given by

- Set of waves up to L_j for every channel s, t, u .

Example $\pi^- \pi^+ \pi^-$: s,t-channel isobars are $\sigma, f_0(L=0), \rho(L=1)$

- Parameterizations of elastic amplitudes $t_l^{(j)}$

Example $\pi^- \pi^+ \pi^-$: $(\pi\pi)_S$ -wave, $(\pi\pi)_P$ -wave from K-function/phase shift

Result of calculation:

- Every isobar rescatters to all others. A solution of the equations tells **a shape and a strength** of the “induced” waves.

Example $\pi^- \pi^+ \pi^-$: Even if f_0 is not coupled to a_1 , it will appear at final state due to rescattering

Input/Output

Model is given by

- Set of waves up to L_j for every channel s, t, u .

Example $\pi^-\pi^+\pi^-$: s,t-channel isobars are $\sigma, f_0(L=0), \rho(L=1)$

- Parameterizations of elastic amplitudes $t_l^{(j)}$

Example $\pi^-\pi^+\pi^-$: $(\pi\pi)_S$ -wave, $(\pi\pi)_P$ -wave from K-function/phase shift

Result of calculation:

- Every isobar rescatters to all others. A solution of the equations tells **a shape and a strength** of the “induced” waves.

Example $\pi^-\pi^+\pi^-$: Even if f_0 is not coupled to a_1 , it will appear at final state due to rescattering

- The solution is linear on production constants, thus we can rescatter every wave **independently**.

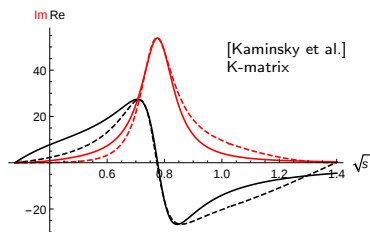
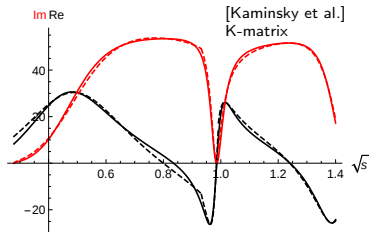
Example $\pi^-\pi^+\pi^-$: final shape and strength of ρ -isobar is

$$A_\rho^{(s)} = (c_\rho a_\rho^{\text{direct}} + c_\sigma a_\rho^{\text{induces}}) P_l(\cos \theta_s)$$

$(\pi\pi)_S - (\pi\pi)_P$ rescattering

Example: $0^- \rightarrow \pi^- \pi^- \pi^+$ system:

- $\pi^- \pi^-$ interaction is negligible.
- $\pi^+ \pi^- (\pi\pi)_S$ $l=0$, i.e. $f_0(500) - f_0(990) - f_0(1500)$, $(\pi\pi)_P$ $l=1$, i.e. $\rho(770)$.
- Simple pole for ρ , three K-matrix poles and background for $(\pi\pi)_S$.

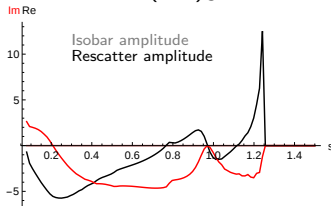
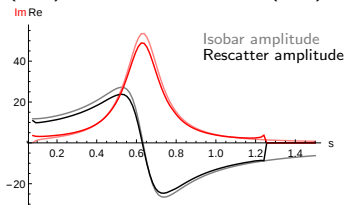


Representation

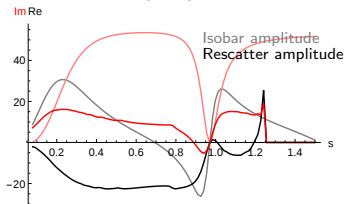
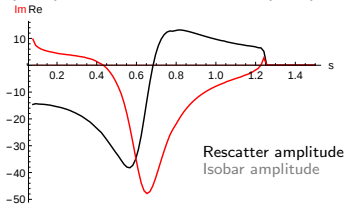
$$A_S^{(s)} = (c_s a_s^{\text{direct}} + c_P a_s^{\text{induces}}) P_0(\cos \theta_s),$$

$$A_P^{(s)} = (c_P a_P^{\text{direct}} + c_S a_P^{\text{induces}}) P_1(\cos \theta_s)$$

- $(\pi\pi)_P$ is a source $\Rightarrow (\pi\pi)_P$ gets modified, $(\pi\pi)_S$ is induced.



- $(\pi\pi)_S$ is a source $\Rightarrow (\pi\pi)_S$ gets modified, $(\pi\pi)_P$ is induced.



Conclusion and outlook

Summary

- Isobar model is a rough approximation, unitarity is not satisfied.
- Rescattering correction can be significant.
- Khuri-Treiman equation gives a tool to deal with a complete series of rescattering.
- KT-PWA analysis is promising extension of PWA.

Prospectives:

- Generalisation of the formalism of the decay of a particle with a spin.
- Implementation of formalism for coupled channels.